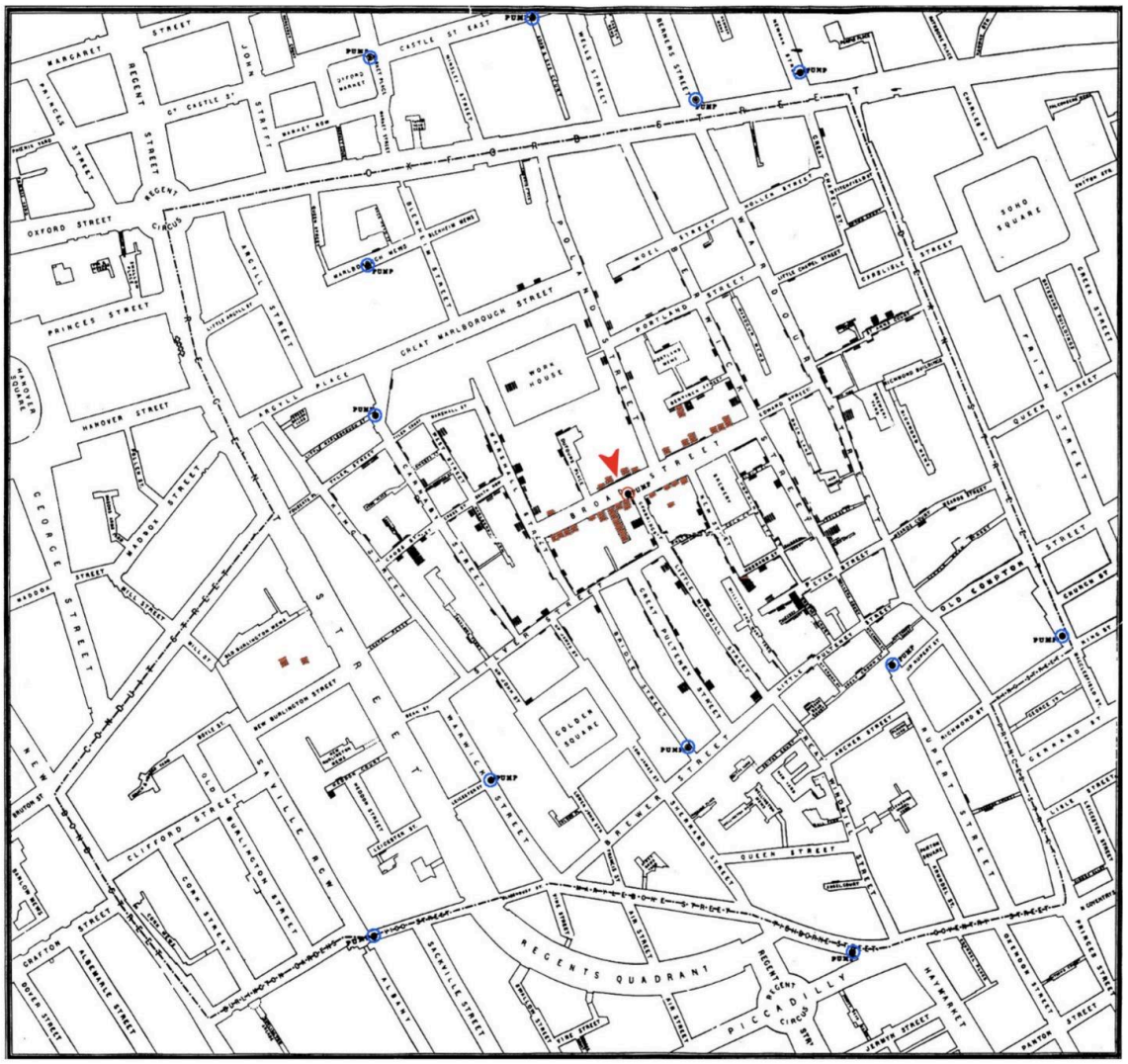
W7 tutorial – Handout materials

This document contains handout materials for the Week 7 tutorial for the AAS 2025/2026 course. The following materials complement presentation slides for the tutorial.

If you have any questions regarding the tutorial, please contact Mikhail Korneev at mikhail.korneev@reuben.ox.ac.uk.

Paper 1 - Snow (1855)

Figure 1.1. Water pumps and the spread of cholera



Paper 2 - Charnysh et al. (2014)

Extract 2.1

At the very beginning of the experiment, each respondent was notified that, as one of the first 100 respondents, she would receive a Rs. 5 bonus. Immediately after, respondents were shown a newspaper report about a fire accident and asked about their willingness to contribute a portion of their bonus to an Indian NGO raising donations for the victims. We primed the salience of Indian national identity by randomly assigning half of the respondents to a treatment condition, in which they were shown a picture of the Indian map shaded in the colors of the Indian tricolor. The image was not displayed to respondents in the control group.

To examine the differences in pro-social behavior toward the ingroup versus the outgroup, we varied the name of the neighborhood in which the fire took place. Half of the respondents were told that a fire broke out in Ramnagar, the other half, in Muzaffarnagar. Ramnagar, as a putatively Hindu name, signals that the victims were likely Hindu, while Muzaffarnagar, a putatively Muslim name, signals that the victims were likely Muslim. Furthermore, we mention specific victims of the fire and vary their names according to the supposed religion of the neighborhood. For the Ramnagar condition, we name Pradeep Kumar, Sunil Raj, and Jaya Kumari as victims. These are deliberately chosen to be 'pan-caste' names that cannot be associated with a

FIRE IN RAMNAGAR.
Times of India, 15 November 2012.

A major fire broke out in *Ramnagar* area of the city on Monday evening. According to officials, over 300 houses and shops were destroyed in the blaze. More than 189 people were injured. There are no reports of casualties though 3 of the injured (*Pradeep Kumar, Sunil Raj, Jaya Kumari*) are said to be in critical condition.

Officials say that the fire began from a shop, most likely due to an electrical short circuit and soon spread to other parts of *Ramnagar*. The incident took place at around 3 pm and the fire could be doused at 6 pm.

We are collaborating with an Indian NGO to raise some donations for the victims of the fire in Ramnagar.

How much of the Rs 5 bonus that you have received would you be willing to contribute to the relief effort?

Figure 1. A screenshot of the experimental page, as seen by a subject receiving the nationalism prime and asked to contribute to victims in the putatively Hindu city "Ramnagar."

specific *jati* or classified as upper or lower caste. For the Muzaffarnagar condition, we mention Abdul Rahim, Fatima Khan, and Ijaz Ahmed. This strengthens the perception of the neighborhood as either Hindu or Muslim. Figure 1 shows a screenshot of one of four experimental conditions (treatment combined with a Ramnagar condition).

The prime and the decision to contribute were followed by a battery of demographic questions. In this section of the survey, respondents were also asked to rank various identities and to tell us how important their Hindu identity was to them. Because the wording of some demographic questions was dependent on answers to others, demographic questions were split into two blocks and randomized within-block. Figure 2 displays a visual map of the experimental flow.

Although the news article was entirely fictitious, respondents were led to believe that the story and the contributions were real. The prompt was designed to mirror real newspaper reports about similar accidents that appear across India almost daily[8] and echoed the commonplace involvement of local NGOs in relief efforts for such accidents. Because we needed to isolate treat-

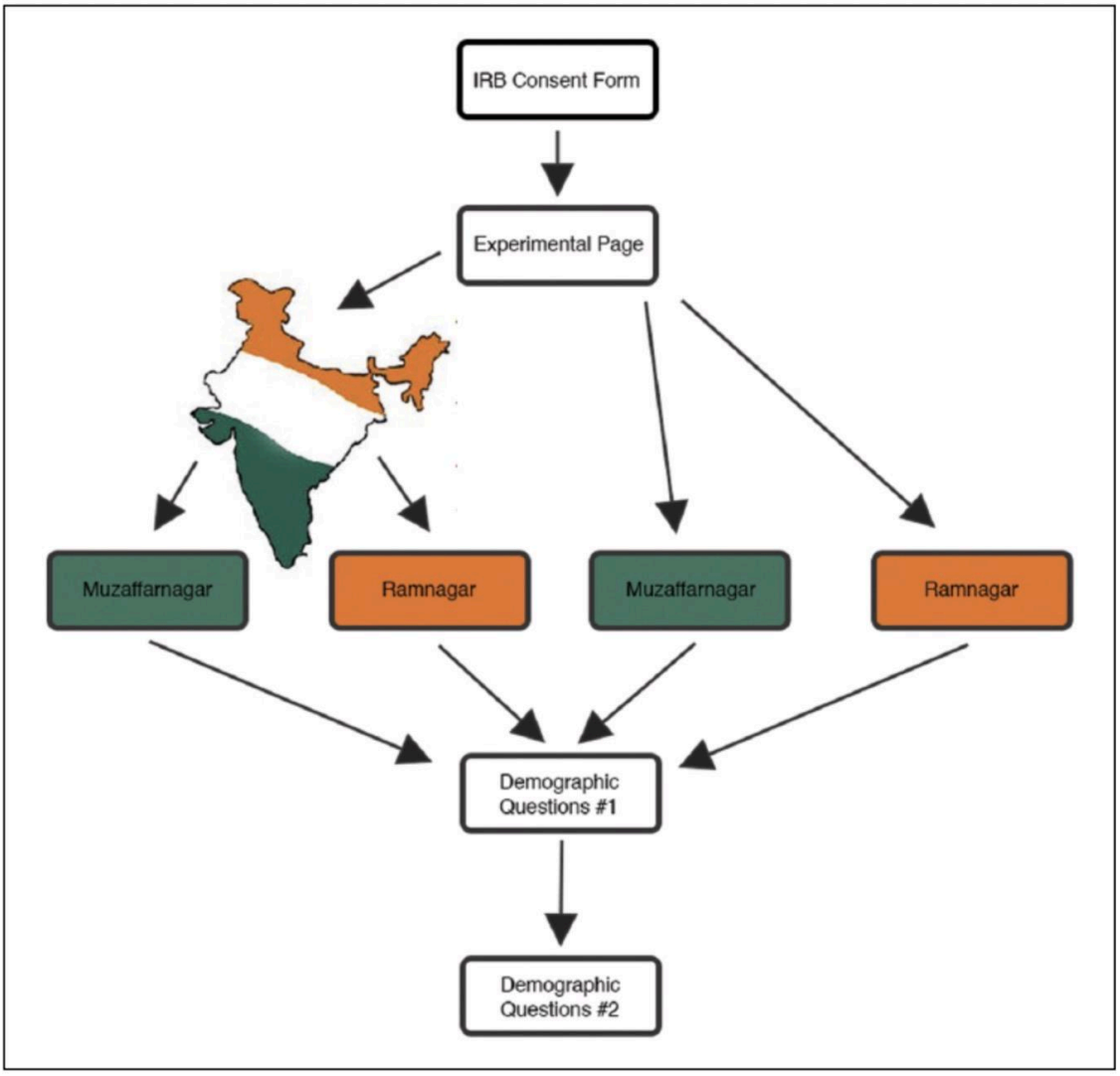


Figure 2. A map of our experimental design.
As displayed, respondents began the survey with a consent form, after which they immediately moved to the experimental portion of the survey. Half of the respondents were shown the prime (the map, shaded in the Indian tricolor), while the other half were shown no such picture (see Figure 1 for a screenshot of this page). After the experimental portion, in which respondents chose to contribute between Rs. 0 and Rs. 5, subjects completed two blocks of demographic questions. Questions were randomized within each of the two demographic blocks. We used two separate blocks because the text of some questions depended on answers to previous questions, and thus we could not randomize across all demographic questions.

Table 2.2. OLS regression outcomes for Charnysh et al. (2014)

Table 3. Regression Results.

	Dependent variable			
	Contribution			
	(1)	(2)	(3)	(4)
Treatment	0.106 (0.115)		0.330** (0.160)	0.381* (0.209)
Ramnagar		0.169 (0.114)	0.403** (0.164)	0.642*** (0.214)
Upper caste				−0.051 (0.233)
Treatment: Ramnagar			−0.455** (0.229)	−0.642** (0.301)
Treatment: Upper caste				−0.116 (0.325)
Ramnagar: Upper caste				−0.553* (0.331)
Treatment: Ramnagar: Upper caste				0.444
Constant	3.012*** (0.082)	2.983*** (0.080)	2.814*** (0.115)	2.834*** (0.149)
Observations	918	918	918	918
R^2	.001	.002	.008	.017
Adjusted R^2	−.0002	.001	.004	.009
Residual *SE*	1.734 (*df* = 916)	1.733 (*df* = 916)	1.730 (*df* = 914)	1.726 (*df* = 910)
F statistic	0.849 (*df* = 1; 916)	2.194 (*df* = 1; 916)	2.342* (*df* = 3; 914)	2.249** (*df* = 7; 910)

*p < .1. **p < .05. ***p < .01.

Paper 3 - Gerber, Karlan, and Bergan (2009)

Extract 3.1. Treatment construction for Gerber, Karlan, and Bergan (2009)

A. *Experimental Procedures*

Households were drawn from a sample of residents in Prince William County, VA, a county 25 miles outside of Washington, DC. The county was suitable for our study because it is within the circulation of both the *Times* and the *Post* but far enough away from the Washington, DC area that the sample is not dominated by citizens involved professionally with politics. We sampled individuals from two lists: a list of registered voters and a consumer database list. Roughly equal proportions were included in the sample from each list.

We performed a baseline survey in September 2005.[3] We asked individuals if someone at the household received either the *Post* or the *Times*, and we excluded from the study those who said they received either newspaper. This perhaps is the most important issue to note regarding the formation of the sample frame. We are studying individuals who do *not* already subscribe to a newspaper, hence are examining the effect of exposing individuals who, on average, are less exposed to the media than the average individual. We also asked a number of other questions about newspaper readership and politics. Only individuals who completed all questions in the initial phone survey were included in the experimental sample.

Individuals were then randomly assigned to one of three groups: the *Post*, the *Times*, or a control group.[4] Participants received a postcard in the mail that said "Congratulations! You have won a free ten-week subscription to the *Washington Times*(*Post*)!"[5] Table 1A shows sample statistics from the baseline interview for the entire sample, broken down by treatment group. The lowest p-value for a test of independence across groups was 0.18 (for gender). Using the treatment assignment as the dependent variable in a multinomial logit model produced a p-value for the joint significance of the covariates of $p = 0.95$.

C. *Outcome Data*

During the week after the November election, we reinterviewed 1,081 of the 3,347 individuals in our sample for the follow-up survey. Response rates of 30 or 40 percent are typical in the public opinion literature (Herbert Asher 2004).[9] Survey respondents were not told of any link between the free subscriptions and the phone surveys. The follow-up survey asked questions about the 2005 Virginia gubernatorial election (e.g., did the subject vote, which candidate was voted for or preferred), national politics (e.g., favorability ratings for Bush, the Republicans, the Democrats, support for Supreme Court nominee Alito), and knowledge of news events (e.g., does subject know the number of Iraq war dead, has subject heard of I. Lewis "Scooter" Libby).

For analyzing the effects on voter turnout, we also obtained administrative records of individual voter turnout and successfully merged these data for 2,571 (76.8 percent) of the individuals in the baseline (not all individuals in the sample frame were registered voters).

Table 3.2. OLS regression results for Gerber, Karlan, and Bergan (2009)

TABLE 4—EFFECT OF TREATMENT ON VOTING BEHAVIOR IN VIRGINIA GOVERNORS RACE
(OLS)

	Voted in 2005 election[a] (1)	Voted in 2005 election[b] (2)	Voted in 2006 election[b] (3)	Voted for Democrat (set to missing if did not vote)[a] (4)	Voted for Democrat (set to zero if did not vote)[a] (5)
Panel A: Separate treatment effects estimated for Washington Post *and* Washington Times					
Washington Post treatment	−0.001	0.011	0.025	0.112	0.072
	(0.033)	(0.019)	(0.019)	(0.045)	(0.035)
Washington Times treatment	0.005	−0.006	0.031	0.074	0.060
	(0.033)	(0.019)	(0.020)	(0.045)	(0.035)
Adjusted R^2	0.21	0.39	0.31	0.31	0.26
F-test: *Post* = *Times*	0.03	0.65	0.10	0.58	0.09
p-value	0.86	0.42	0.75	0.44	0.76
Panel B: Pooled treatment effect estimated for receiving either newspaper					
Received either *Post* or *Times* treatment	0.002	0.003	0.028	0.093	0.066
	(0.028)	(0.016)	(0.016)	(0.038)	(0.029)
Adjusted R^2	0.21	0.39	0.31	0.31	0.26
Observation counts for both panels					
Observations	1,079	2,571	2,571	718	1,003
Refused/not asked	2			363	78
Total not merged (columns 2 and 3)		776	776		
Total surveyed in follow-up	1,081			1,081	1,081

Notes: Standard errors in parentheses. The following covariate variables are included in all specifications: gender; reported age; three separate indicators for voting in the 2001, 2002, and 2004 general elections; an indicator for whether the respondent was drawn from a consumer list; self report of receiving any news or political magazines; baseline survey self reports of preferring the Democratic candidate in the gubernatorial election and having no preference in the gubernatorial election; and an indicator for wave of the study. If a covariate value was missing, an indicator variable was included and the covariate was coded as zero. We include strata indicators, which are variables for each strata formed prior to the randomization, which included unique combinations of the following: intention to vote, receive a paper (non-*Post*/non-*Times*), mentions ever reading a paper, gets a magazine, and asked whether they wish they read the paper more. All results remain qualitatively similar, and statistical significance remains as-is, using probit specifications instead of OLS.

Data source:

[a] Survey.

[b] Administrative voting records.